

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

ERROR CORRECTING CODES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) How are channels classified ? Give characteristics of Discrete Memoryless Channel. 4(CO1)

(b) A DMS has six symbols $x_1, x_2, x_3, x_4, x_5, x_6$ with $P(x_1) = 0.35, P(x_2) = 0.25, P(x_3) = 0.2, P(x_4) = 0.1, P(x_5) = 0.06, P(x_6) = 0.04$:

(i) Construct a Shannon-Fano code for X. Calculate efficiency of the code.

(ii) Repeat for Huffman code and compare the results. 6(CO1)

2. (a) Elaborate Go Back N ARQ and Selective Repeat ARQ Systems. 4(CO2,3)

(b) The parity check matrix of a particular (7, 4) linear block code is given by :

$$[H] = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

(i) Find the Generator Matrix G.

(ii) List all code vectors.

(iii) Comment on how many number of errors can be detected and corrected. 6(CO2,3)

3. (a) For systematic (7, 4) cyclic code, determine the generator matrix and parity check matrix. Given : $G(p) = p^3 + p + 1$. 7(CO2,3)
- (b) Discuss in brief about special features of cyclic codes. How systematic and non-systematic codes are generated in cyclic codes ? 3(CO2,3)
4. (a) A convolution encoder is described by
 $g_1 = [1 \ 1 \ 1]$, $g_2 = [1 \ 0 \ 0]$, $g_3 = [1 \ 1 \ 0]$
 Draw :
 (i) The encoder corresponding to this code.
 (ii) The code tree for this code.
 (iii) The state diagram for this code.
 (iv) The trellis diagram for this code.
 Decode following received sequence using viterbi decoding method :
 $Y = 110111001$ 10(CO5)
5. Analyse Galois fields in RS codes with an example. Also formulate structure of a typical RS code. 10(CO4)
6. (a) Discuss encoding and decoding process of Turbo codes with one example. 5(CO4)
- (b) Illustrate LDPC codes with an example. Specify the merits and demerits of LDPC codes. 5(CO4)

